

Artificial Intelligence for Corporate Finance

AI-Augmented Analyst Workflows with Excel, Python and LLMs

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Structure of the course

- **Email:** amedeo.andriollo@dauphine.psl.eu
- **Assessment:** 150-minute final in-class examination.
- **Office hours:** feel free to (DO) come: B612.
- Slides/materials are updated regularly. Report a typo/mistake: corrections help the class.
- **This session:** a first-pass sell-side target study before valuation, with AI as augmentation.
- **Key references:** i) Rosenbaum & Pearl, Chapter 1 “Study the Target” (Exhibit 1.3); ii) official OpenAI and Anthropic guidance; iii) Anthropic AI Fluency.

Learning objectives

During your internship...

Before a sell-side team values a company, it turns a compact, incomplete pack into a **concise view** of the business, operating history, unresolved tensions and next questions.

Today's learning sequence

1. Study TargetCo without a prompting method.
2. Compare outputs and (partial) feedback.
3. LLM and disciplined workflow.
4. Redo the task in a fresh chat, then inspect the full answer and workflow.

Structure of our course: target-study note → audited Excel workbook/change log → valuation analysis → reconciled Python/Excel analysis → checked deal-extraction table + derived transaction indicators → strategic buy-side briefing/workflow.

Study the target before choosing how to value it

The finance operation

“Study the Target” means learning the company’s story before selecting or benchmarking comparables. A compact target study is neither due diligence nor valuation. It creates the working frame for later choices.

Business profile

- sector and subsector;
- products and services;
- customers and end markets;
- distribution channels;
- geography.

Financial profile (selected dimensions)

- size;
- profitability;
- growth;
- (returns and credit profile later).

Source basis: Rosenbaum & Pearl (2013), Ch. 1, Step I “Study the Target” (Exhibit 1.3).

Reference

Simple measures answer different economic questions

Revenue growth and its disclosed bridge

$$g_t = \frac{R_t}{R_{t-1}} - 1, \quad g_t = v_t + p_t.$$

Here v_t and p_t are disclosed volume and price/mix contribution points to revenue growth.

Question answered: how did sales change, and which disclosed contribution drove the change?

Reported EBITDA margin

$$m_t = \frac{\text{EBITDA}_t}{R_t}.$$

Question answered: how much operating profit is reported per euro of revenue?

Growth vs. margin: they can move differently, as margin also depends on cost recovery, utilization, mix and other operating factors.

TargetCo: 2025A utilization exceeds 2022A, while the 2025A reported margin is lower.

↪ analytical tension (not clear causality).

Baseline design: today's first assignment

	ChatGPT group	Claude group	Human-only group
Inputs	Same two-page TargetCo profile; same VP assignment; same time.		
Baseline	Student-composed 1 message prompt from common sketch.	Same finance assignment.	Same finance assignment w/out AI.
Redo	Temporary chat (max 3 messages).	Incognito chat (max 3 messages).	Possible AI access.
Main comparison	Baseline output/work process versus redo output/work process		

Interpretation discipline

The baseline is not a ChatGPT vs. Claude comparison: students may formulate different prompts.
↪ different student/AI workflows under a common finance assignment, material, time and prompt budget.

Baseline prompt sketch

AI groups: one message, with student freedom

You may edit this basic starting sketch before sending your one message.
The VP request remains the separate finance assignment.

*Using the supplied TargetCo profile, help me prepare the one-page note requested by the VP.
Use the supplied material as the source. Do not value the company.*

Human-only group: complete the same finance assignment from the same profile, without AI.

Record: retain the exact prompt your group actually used: NO second message allowed.

20-min baseline: produce a first-pass VP note

Deliverable

One page. Use only the supplied material. Do not value TargetCo.
AI groups retain the actual conversation; human-only groups retain notes.

Read for

- business configuration;
- commercial facts and operating history;
- what is known and what is missing.

Do not do

- external research;
- peer selection or valuation;
- a generic diligence inventory;
- a prompting-method exercise.

Start now. We compare outputs in about 20'.

A stronger single-shot instruction

Instructor example: still one message

Use only the supplied TargetCo profile as the permitted source. Prepare one concise, one-page first-pass target-study note for the VP before valuation. Do not use external facts and do not value the company.

1. **Business model / positioning:** synthesize products, qualification, customers/end markets, direct/distributor route to market and geography. State only what the profile supports.
2. **Historical trajectory:** recompute revenue growth where possible. Treat volume and price/mix as disclosed contribution points to revenue growth. Interpret utilization and reported EBITDA margin without asserting an unproven causal bridge.
3. **Key tensions:** identify the material evidence gaps or management statements that require qualification.
4. **Three ranked questions + why each matters:** give exactly three questions. For each, name the later judgement it could change: revenue, margin, normalization, risk, peer selection or valuation.

For each material number or management statement, use a brief profile/table locator. Keep observed data, management statements and analyst inferences distinct.

Source basis: Spoiler of the full sequence (three messages).

Reference

Four observable dimensions for the VP note

Dimension	Score 2/2:
Business-model synthesis	Connects product specification, customer/end-market mix, direct/distributor route to market and geography/footprint into a coherent operating configuration. It does not invent pricing power.
Historical economics	Reconciles revenue growth; distinguishes volume contribution from price/mix contribution; interprets utilization and reported EBITDA-margin movement; identifies association rather than unproven causality. Numerical fidelity belongs here.
Certainty vs. Uncert.	Keeps observed data and management statements distinct; qualifies causal interpretation; imports no external facts; turns unsupported statements into questions where appropriate.
VP usefulness	Uses exactly three ranked questions. Each says which later judgement could change. The note is concise and decision-useful.

Rate yourself: **0** absent, incorrect; **1** present but partial, or mechanically copied; **2** meets standards.

Source basis: Common rubric for baseline, redo and optional homework.

[Reference](#)

Partial answer: an analytical route

Route

1. Build a synthesis of the operating model.
2. Explain the historical change in drivers, rather than list annual snapshots.
3. Name tensions created by the supplied doc.
4. Ask only questions whose answers could change a later valuation judgement.

Yet to come:

- complete model answer;
- complete ranked-question set;
- full fact/management/inference treatment;
- multi-turn instructor workflow;
- final source reconciliation.

The key for a good starting point:

Make a small evidence pack decision-useful while preserving what it does not establish.

Working interpretation about revenues

	2022A	2023A	2024A	2025A
Revenue growth	–	2.1%	1.6%	5.9%
Volume contribution (ppt)	–2.5%	–4.0%	0.5%	4.0%
Price / mix contribution (ppt)	10.0%	6.1%	1.1%	1.9%
Utilization	82%	78%	74%	84%
Reported EBITDA margin	14.8%	14.0%	12.8%	14.3%

How to read the table. Positive price/mix contribution can offset falling volume. Recovering utilization can coexist with incomplete margin recovery. The table therefore shows co-movement, not a causal margin!

A defensible reading

In 2023A, +6.1 ppt price/mix offset a –4.0 ppt volume contribution.

In 2025A, +4.0 ppt volume is the larger disclosed contributor.

Utilization recovered to 84%, but the 14.3% margin remains 50 bp below 2022A.

Recovery is evident; the supplied pack does not quantify the margin bridge or prove a steady-state outcome.

Source basis: Full commentary.

Reference

Partial feedback: business-model synthesis

What the pack supports

TargetCo is a specification-led manufacturer rather than merely a reseller. Engineering, tooling and customer qualification, combined with a 71% direct/key-account mix, indicate a service- and qualification-intensive route to market. The distributor channel and European footprint add commercial context.

What it does not support/establish

It does not establish pricing power, retention, channel margin or contract durability. The 9.4% largest-customer and 41.8% top-ten figures are facts requiring context, not an automatic risk classification.

Valuation-relevant question

What is the end-market, channel and facility-level evidence behind the 2025A volume recovery?

Its answer could change revenue confidence, risk and later peer selection.

Source basis: What the two-page Target Co profile can support

Reference

LLMs generate conditional text, not truth

Term	Definition
AI	Computational systems that perform tasks associated with perception, prediction, generation or decision support.
Generative AI	AI that produces new content, such as text, code, images or audio.
LLM	A generative model trained to predict and generate sequences of language tokens.
Token	A unit of text processed by the model; a token is not reliably a word.
Context window	The finite token sequence the model can use at one time: instructions, documents, prior messages and current task instruction.
Inference / generation	Running the already-trained model: given current context and tokens already generated, the model computes a conditional distribution over the next token; the decoding rule selects or samples a token and the process repeats until completion.

Analytical implication

An answer can be grammatically convincing while misstating a source, skipping a number or inventing a causal story (e.g., check "AI allucination"). Language quality is not source fidelity.

Source basis: Model, Application Tool in workflow

Reference

Autoregressive generation explains why context changes outputs

A minimal technical model

$$p_{\theta}(x_1, \dots, x_T \mid c) = \prod_{t=1}^T p_{\theta}(x_t \mid x_{<t}, c).$$

- x_t is the next token; $x_{<t}$ is the preceding generated sequence.
- c is supplied context: task, source material, examples and prior conversation.
- θ denotes learned model parameters. At generation time, the application repeatedly chooses or samples a next token from a conditional distribution.

Consequence

Changing a prompt or context changes the distribution of possible outputs. It does not create a criterion for whether an output is true, financially reasonable or properly sourced.

Context design: evidence, authority and salience

A bit of terminology:

Evidence: information from the (restricted) source used to support a claim/calculation.

Source authority: the status assigned by the analyst/task specifying which source governs a claim when sources differ.

Salience: how strongly a supplied item effectively influences generation.
Analyst importance does not imply model salience.

TargetCo contrast

The profile is **authoritative** for this exercise. M-01 and M-02 inside it remain **management statements**, not independently established facts.

More context can add evidence but also irrelevant instructions, competing sources, weak salience and verification cost. Identify the permitted source, preserve its boundary, then request extraction before synthesis.

Source basis: Long-context failures

Reference

Model vs. Application vs. Tool

TargetCo PDF → application handles/parses file and manages chat/context → underlying model generates response → calculator, code or file tool may retrieve, recompute or check → analyst reviews result

Object	Meaning in this pathway
Model	The underlying generative model that produces conditional token predictions (e.g., Codex 5.6 Terra).
Application	The interface/orchestration layer that may handle files, memory, system instructions, context and model routing (e.g., Claude code vs. Claude browser chat).
Tool	An external capability, such as a calculator, code execution or file operation. It can retrieve or recompute something, creating a separate source and checking.

Source basis: Model, Application Tool in workflow

Reference

AI fluency = human work before, during and after AI use

Anthropic's 4D framework

Delegation – decide whether/when/what to hand to AI.

Description – communicate goal, context and standards.

Discernment – assess quality and usefulness.

Diligence – take responsibility for use and final delivery.

Modes

Automation: AI performs a routine task.

Augmentation: AI expands a person's analysis while the person directs and judges it.

Agency: AI pursues a multi-step objective with delegated action scope.

Course map: Session 1 is principally **augmentation**: delegate extraction and drafting, retain judgement and verification.

(Session 6 moves closer to agency, with plans, approval points and stopping rules.)

Source basis: Session 1 and AI fluency 4Ds

Reference

Define success before improving the prompt

Three different objects

Success criterion = a property the output should satisfy. **Evaluation** = an observable test of that property.
Failure signal = evidence that the criterion was not met.

Criterion	Observable test	Failure signal
Business-model synthesis	Does the answer connect product/qualification, customers and end markets, route to market and footprint?	Copies profile headings, calls TargetCo a reseller or invents pricing power.
Historical economics	Does it reconcile revenue growth, distinguish volume from price/mix contribution points and interpret utilization/margin cautiously?	Calls 2023A volume-led growth or asserts an unsupported causal margin bridge.
Evidence discipline	Are source facts, management statements, calculations and analyst inference kept distinct?	Reports M-02's "non-recurring" characterization as independently established fact.
VP usefulness	Are there exactly three ranked questions, each linked to a later judgement?	Produces a generic, unranked diligence list.

Principle: define what "good" means before trying to make the prompt more elaborate.

Source basis: Evaluation comes before prompt cleverness.

Reference

From baseline prompt to reviewable workflow

Stage	Design choice added	Problem it is meant to address
P0	Baseline student prompt	Establish what happens under the common assignment without an advanced prompting method.
P1	Specify source, task and output	Reduce task drift: authoritative source, four sections, exactly three ranked questions, no valuation.
P2	Make evidence and calculations visible before drafting	Prevent polished prose from hiding an extraction, arithmetic or classification error.
P3	Attach explicit criteria and checks	Make quality testable rather than relying on “looks good.”
(P4)	Separate the work into stages	Localize errors and create points at which the analyst can intervene.

Important

P4 summarizes the shift from a single, lengthy prompt to a structured workflow:

Extract & analyse → Challenge → Draft → Verify & reconcile.

Source basis: Prompt-chain skeleton.

[Reference](#)

P1 → P2: expose the work before the prose

P1: strong single-shot specification

Use only the supplied TargetCo profile.

Produce:

- business model / positioning;
- historical trajectory;
- key tensions;
- exactly three ranked questions + why.

No external facts. No valuation.

P2: first produce a reviewable artifact

Before drafting:

- extract business-model evidence;
- recompute revenue growth;
- identify volume and price/mix contribution points;
- track utilization and reported margin;
- classify material statements;
- state what each item supports and does not establish.

What changed? P1 improves the specification. P2 makes the evidence, calculations and classifications available for review *before* they are compressed into prose.

Source basis: Prompt-chain skeleton.

Reference

P3 → P4: test the work, then separate the stages

P3: explicit evaluation

Before accepting the answer, test:

1. business-model synthesis;
2. historical economics;
3. evidence / uncertainty discipline;
4. VP usefulness and prioritization.

Also reconcile material numbers and claims.

P4: a reviewable chain

Extract & analyse



Challenge



Draft



Verify & reconcile

Trade-off

Benefit: errors become easier to locate and humans can intervene (divide et impera).

Cost: more interactions, context/state management and possible inconsistency across stages.

Source basis: Prompt-chain skeleton.

[Reference](#)

Three prompt devices and their limits

Device	Useful when...	It cannot...
Examples	The desired output form is ambiguous (e.g., concise phrasing or question structure).	Supply TargetCo evidence or make an unsupported claim true. A filled TargetCo example before baseline may also leak the answer.
Structure	A prompt contains several components, which might be separated: task, sources, output, criteria and checks (see next slide).	Validate arithmetic or compensate for missing evidence.
Role / audience	Tone, vocabulary and decision context matter: e.g. "write for the VP using the supplied profile."	Create privileged information, prove causality or transfer accountability.

Common misunderstanding

Prompt devices shape how the task is communicated.
They do not add evidence and they do not perform independent verification.

Source basis: Examples of Appropriate vs. Inappropriate.

[Reference](#)

A strong prompt makes five things visible

Portable structure

Task	Prepare a one-page first-pass TargetCo note for the VP before valuation.
Sources	Use the supplied TargetCo profile only. No external company facts.
Output	Business model / positioning; historical trajectory; key tensions; exactly three ranked questions + why.
Criteria	Coherent operating configuration; correct historical economics; evidence discipline; VP usefulness.
Checks	Recompute the revenue bridge and reconcile every material number or statement to source or calculation.

Formatting is secondary. Headings, XML tags or another delimiter can express the same structure.

↪ The important object is the semantic separation of task, source, output, criteria and checks.

Source basis: The same semantic prompt in headings and XML.

Reference

Verification ladder: from form to evidence

Level	Question	TargetCo example
1. Form / coverage	Did the output actually satisfy the assignment?	Four sections; exactly three ranked questions; one-page form.
2. Source reconciliation	Does every material claim return to the permitted source?	Confirm EUR 481.0m revenue and preserve M-02 explicitly as a management statement.
3. Independent audit or recomputation	Can a result be checked independently of the generated prose?	Starting from the source inputs, recompute $481.0/454.0 - 1 = 5.9\%$ in a calculator or spreadsheet.

Boundary

Asking the same model “Are you sure?” can be a useful self-check. It is **not** an independent verification.

Source basis: Self-check vs. independent check.

Reference

Classify each material claim before synthesis

1. Source fact

Something directly reported in the permitted source.

Example: 2025A utilization = 84%.

2. Management statement

Something management asserts, but which the pack does not independently establish.

Example: M-02 describes 2024A–2025A commissioning and duplicate-quality costs as “non-recurring.”

3. Analyst calculation

A mechanically reproducible transformation of source inputs.

Example:

$$481/454 - 1 \simeq 5.9\%.$$

4. Analyst inference

A qualified interpretation requiring judgement.

Example: operating recovery is evident, but the supplied pack does not establish a steady-state margin.

Different checking obligations: facts return to source; calculations are recomputed; management statements remain attributed; inferences remain qualified.

Evidence matrix: what we know and where we stop

Analytical point	Evidence / source	Supports	Does not establish
2023A growth character	Historical table: volume -4.0 ppt; price/mix +6.1 ppt; revenue growth +2.1%.	Price/mix more than offsets the disclosed negative volume contribution.	Customer-level pricing, mix or volume drivers.
2025A utilization / margin tension	2022A vs. 2025A: utilization 82% → 84%; reported margin 14.8% → 14.3%.	Higher utilization alone cannot explain the complete margin outcome.	A quantified causal narrative across utilization, mix, cost and facility effects.
M-02 normalization question	Management labels commissioning and duplicate-quality costs in 2024A–2025A “non-recurring.”	Cost nature and duration are valuation-relevant diligence questions.	That the costs are recurring, non-recurring, or that any particular adjustment is justified.

Purpose of the matrix: preserve the boundary between **evidence**, **supported interpretation** and **what remains unsupported**.

Source basis: Full(er) evidence matrix.

Reference

Redo: same assignment, different process

Start clean

Use a **fresh chat** with the same TargetCo profile and the same finance assignment. The objective is not to erase a bad answer. It is to compare two different workflows under the same underlying task.

Maximum three messages

1. **Extract & analyse**
Evidence, calculations, classifications, support and limits.
2. **Challenge + draft**
Apply the four criteria, correct supported problems, then draft.
3. **Verify & reconcile**
Return every material number and claim to source or calculation; revise only where justified.

3 messages, 4 logical stages: Extract & analyse → Challenge → Draft → Verify & reconcile.

Record: prompts used; what was delegated or not; one challenged claim; one numerical/source check; one remaining doubt.

Post-redo answer I: synthesis, not inventory

Business model / positioning

TargetCo is a European specialty manufacturer of technically specified rigid containers, closures and dispensing assemblies. Engineering, tooling and customer qualification are part of the offer; 71% of FY2025A revenue comes through direct/key accounts and 29% through selective distributors. Food & Beverage is the largest disclosed end market (55%). These facts support a specification- and service-intensive operating configuration, but the pack does **not** establish pricing power, retention or channel economics.

Historical trajectory

Revenue increased from EUR 438.0m in 2022A to EUR 481.0m in 2025A. In 2023A, +6.1 ppt price/mix offset a -4.0 ppt volume contribution, producing 2.1% growth. In 2025A, the larger contribution is volume: +4.0 ppt versus +1.9 ppt price/mix. Utilization recovered to 84%, above 2022A's 82%, while reported EBITDA margin remained lower at 14.3% versus 14.8%. The recovery is visible; the supplied pack does not provide the full margin bridge.

What makes this synthesis? The answer connects facts into an operating story while preserving what the evidence does not establish.

Source basis: Full(er) evidence matrix.

Reference

Post-redo answer II: tension → question → judgement

Evidence tension	Ranked question	Later judgement affected
2025A utilization is 84% versus 82% in 2022A, yet reported margin is 14.3% versus 14.8%.	1. What quantitatively bridges the 2022A–2025A margin gap, and what evidence supports a sustainable run-rate?	Margin forecast; valuation.
M-02 labels costs appearing in both 2024A and 2025A “non-recurring.”	2. What is the nature, amount and duration of the commissioning and duplicate-quality costs?	Normalization judgement.
2025A volume contribution is +4.0 ppt, but the pack gives no customer/channel/end-market/facility bridge.	3. How broad and durable is the volume recovery across customers, channels, end markets and facilities?	Revenue confidence; risk; peer selection.

Evidence boundary

M-01 and M-02 remain management statements. The pack supports questions about recovery and normalization: neither forecast nor valuation.

Source basis: Why the three questions are ranked.

Reference

AI contributes work; the analyst retains judgement

Stage	AI can assist with	Analyst must own
Extract & analyse	Locate evidence, organize facts, recompute arithmetic, propose classifications.	Which source is authoritative; whether the extraction is complete and correct.
Challenge	Surface contradictions, omissions, weak questions and unsupported language.	Whether causal language is justified and which tensions are economically material.
Draft	Compress the reviewed analysis into VP-ready prose.	Prioritization, professional judgement and communication standard.
Verify & reconcile	Produce a reconciliation list and identify remaining inconsistencies.	Independent checking, normalization decisions and final acceptance.

Connection to the 4Ds

Delegation: what AI does. **Description:** task/source/criteria. **Discernment:** challenge and evaluate.
Diligence: verify and own the final deliverable.

Source basis: Full prompt in SESSION1_INSTRUCTOR_AI_WORKFLOW.md.

Reference

Take some time to compare the actual outputs

Baseline versus redo

- Which answer correctly decomposes the historical revenue story?
- Which recognizes the utilization/margin tension?
- Which question would change later valuation assumptions most?
- Which output reports a management assertion as if independently verified?
- Which output invents a causal explanation unsupported by the pack?

Optional homework: same-prompt

cross-application comparison Use the same instructor near-final prompt sequence, same TargetCo profile and fresh Temporary/Incognito chats. Match tools, score both on the common rubric, record date/application/displayed model-version/tool-memory state, then write what differences matter to a VP.

The work product changes; accountability does not

Session	Professional work product
1	Target-study note.
2	Audited Excel workbook and change log.
3	Valuation analysis.
4	Reconciled Python/Excel analysis.
5	Checked deal-extraction table plus derived transaction indicators.
6	Strategic buy-side briefing/workflow.

What changes — and what does not

The finance artifact, technical environment and permissible degree of AI delegation become progressively more demanding.

The analyst still owns specification, evidence, judgement and verification.

Before Session 2: from narrative to model audit

Next time

Can we trust TargetCo's historical Excel model?

Refresh the finance concepts

- reported vs. adjusted EBITDA;
- debt, cash, net debt and leverage;
- CapEx and NWC consistency;
- formula tracing and one-time / non-recurring items.

Reading

- Rosenbaum & Pearl, Ch. 1: Financial Profile and Calculation of Key Financial Statistics and Ratios.
- Workbook Ch. 1: Q3, Q7–9 and Q14–21.
- Optional: Koller, Goedhart & Wessels (2025), Chs. 11 and 13, for model organisation/hygiene.

The task changes from **interpreting a supplied narrative pack** to **inspecting and independently checking model logic**.

The analyst owns the specification and the check

Today's lesson

A strong target study turns a small evidence pack into a concise view of the business, operating history and valuation-relevant uncertainty.

AI can accelerate that work. It cannot decide which evidence governs, whether a causal interpretation is justified or whether the final deliverable is acceptable.

Extract & analyse → Challenge → Draft → Verify & reconcile

Before next session

Keep the final AI Work Record and prompt-chain locator.

In Session 2, the question becomes: **can the historical Excel model be trusted after inspection and independent checks?**

Appendix: target-study profile and later valuation work

Business profile: why each dimension matters

- Sector/subsector frames drivers, cyclicity, etc.
- Products/services distinguish commodity-like supply from specified/service-intensive offers.
- Customers/end markets distinguish purchaser from demand exposure.
- Channels/geography can alter commercial model, cost base and comparability.

Financial profile: why selected dimensions matter

- Size can correlate with scale and market comparability.
- Profitability describes conversion of sales into profit.
- Growth needs a driver-level reading.
- Returns/credit require later calculations.

Source basis: Rosenbaum & Pearl (2013), Ch. 1, Step I and Exhibit 1.3; original paraphrase.

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Appendix: what the two-page TargetCo profile can support

Evidence supplied	A cautious analyst may say	A cautious analyst must stop short of saying
End-market, geography, channel and concentration percentages	Reported mix and configuration follow-up.	Concentration is high/low, commercially good/bad or contractually secure.
Revenue bridge, utilization and margin history	2023A price/mix contribution offsets falling volume; 2025A has a larger volume contribution; utilization and margin recover from 2024A.	A quantified causal connection, or sustainable run-rate.
M-01 and M-02 management statements	What management says about recovery levers and cost classification.	That recovery will occur or an adjustment is justified.

Classroom purpose

The analyst must synthesize, calculate, challenge and prioritize.
The short deck is deliberately incomplete!

Source basis: Back.

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Appendix: four-dimensional rubric calibration

Dimension	0	1	2
Business-model synthesis	Copies labels or omits configuration.	Names facts but weakly connects them.	Connects specification/qualification, customer/end market, route to market and footprint without inventing economics.
Historical economics	Wrong contribution story, arithmetic or causality.	Lists figures with a weak bridge.	Reconciles growth, distinguishes contributions, treats utilization/margin cautiously and keeps numerical fidelity.
Evidence and uncertainty discipline	Imports facts or treats management assertion as fact.	Notes uncertainty but blurs categories.	Separates observed data, management statements and qualified inference; turns unsupported claims into questions.
Prioritization / VP usefulness	Not three questions or generic list.	Three questions but weak consequence or unclear note.	Exactly three ranked questions; each names a later judgement; concise VP-ready note.

Source basis: Back.

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Appendix: model, application and tool in the TargetCo workflow

Concrete pathway

TargetCo PDF → application parses the file and manages the chat/context → model generates the response → a calculator/code/file tool may retrieve, recompute or check a result → analyst reviews it.

Term	Operational consequence
Model	Produces conditional next-token predictions. Record any displayed model/version; do not infer hidden configuration.
Application	May add file conversion, system instructions, context management, routing, memory or tool access. An application comparison is not a pure model experiment.
Tool	Retrieves, calculates or changes an external object. Tool output creates a source and checking obligation.

Source basis: Back.

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Appendix: long-context failure modes

Possible drawbacks

- management material and an authoritative source are treated as equally established;
- source material contains instructions that compete with the task;
- material evidence is lost among irrelevant pages;
- a citation label appears without a reconciled claim.

Fixes

- label source, date and authority;
- delimit documents and permitted use;
- make the task instruction easy to locate;
- extract fact/statement/calculation with location before synthesis.

TargetCo lesson

The profile is authoritative for the exercise while M-01/M-02 within it remain management statements. Authority and truth status are different analytical questions.

Source basis: Back.

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Appendix: evaluation comes before prompt cleverness

Evaluation loop

task-specific test cases → initial prompt → run → score → diagnose → revise.

Observed failure	Diagnosis	How to fix
Calls 2023A demand/volume-led	The output ignores disclosed −4.0 ppt volume and +6.1 ppt price/mix contributions.	Require source-grounded extraction and contribution recomputation before prose.
Treats M-02 as fact	Management-statement classification/source-fidelity check is missing.	Request classification, locator and analyst verification before drafting.
Gives six generic questions	Prioritization criterion is not operational.	Require exactly three ranked questions and one named valuation consequence each.
Correct but too long	Output constraint fails.	State one-page constraint and use a non-TargetCo output-shape example if format remains unstable.

Source basis: Back.

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Appendix: examples as prompt devices

Non-TargetCo output-shape example

Observed fact: a product requires customer qualification before serial production.

Restrained interpretation: qualification may make switching and implementation relevant to diligence.

Open question: what evidence shows the duration, economics and renewal pattern of qualified relationships?

Appropriate Use this type of example *after baseline* when a group needs a model of fact → qualified interpretation → question.

Inappropriate before baseline A filled TargetCo example leaks the analytical answer, anchors wording and turns the task into transcription.

Source basis: Back.

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Appendix: the same semantic prompt in headings and XML

Neutral headings

Task

Prepare a one-page TargetCo VP note.

Sources

Supplied TargetCo profile only.

Output

Four named sections.

Criteria

Four observable success criteria.

Checks

Recompute bridge; reconcile claims.

The tags separate task, source boundary, requested output, criteria and checks. They do not create a different reasoning method or independently verify a result.

Anthropic XML

```
<task>Prepare a one-page TargetCo VP note. </task>
```

```
<sources>Supplied TargetCo profile only. </sources>
```

```
<output>Four named sections. </output>
```

```
<criteria>Four observable success criteria. </criteria>
```

```
<checks>Recompute bridge; reconcile claims. </checks>
```

Source basis: Back.

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Appendix: Roles set audience and standards

Prompt A

You are a senior investment banker. Explain TargetCo's prospects and give a valuation view.

Problem: the role conflicts with the assignment, invites external knowledge and encourages an unsupported valuation claim.

Prompt B

Write for a VP considering a sell-side mandate. Use only the supplied profile. State the operating picture, label management statements and leave unsupported causality as questions. Do not value the company.

Benefit: audience, vocabulary, source boundary and standards are explicit.

Rule

A role guides tone, context and perspective. It cannot add facts, make an inference true or transfer accountability.

Source basis: Back.

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Appendix: prompt-chain skeleton

Stage	Output	Human question	Failure localized
Extract	Evidence table; calculations	Right facts, contribution points and source locations?	Retrieval / arithmetic
Challenge	Criteria-based corrections	What is unsupported, omitted or generic?	Interpretation / specification
Draft	One-page VP note	Is it concise, qualified and decision-useful?	Communication / prioritization
Verify	Reconciliation log	Does each material point return to a source or calculation?	Source fidelity

State-management cost

Each stage needs a stable source pack, versioned prompt and handoff.

Source basis: Back.

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Appendix: Self vs. independent check

Check request	What it can reasonably do	What is still required
"Review this draft against the rubric."	Detect missing sections, generic questions or an obvious internal contradiction.	Reconcile each material statement to the supplied profile.
"Check the calculations."	Recalculate a percentage or identify arithmetic inconsistency.	Independently recompute in calculator/spreadsheet and identify inputs/source.
"Are you sure M-02 is non-recurring?"	Surface qualification language or admit missing support.	Obtain evidence on cost nature/duration; analyst decides later normalization treatment.

TargetCo implication

M-02 spans two years. That supports a question about recurrence and duration; it neither proves recurrence nor non-recurrence, and it establishes no adjustment amount.

Source basis: Back.

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Appendix: fuller TargetCo evidence matrix

Analytical point	Source location	Evidence/calculation	Support, limit and next question
Specification-led configuration	Company extract; product/facility notes	Qualification before serial production; containers, closures and dispensing components; 71% direct/key account mix	Supports specialty manufacturing/service configuration; does not show pricing power or channel margin.
2023A growth character	Historical table; 2023A commentary	Volume contribution −4.0 ppt; price/mix +6.1 ppt; revenue +2.1%	Supports contrast with 2025A; no customer-level price/mix bridge.
2025A utilization/margin tension	2022A–2025A table	Utilization 82% to 84%; margin 14.8% to 14.3%	Supports challenge to single-driver causality; open cost/mix/facility bridge.
Customer concentration	Commercial snapshot	Largest 9.4%; top ten 41.8%	Supports request for contract and retention context; no automatic risk conclusion.
M-01/M-02	Management materials	Asserted recovery levers; two years of stated non-recurring costs	Supports diligence questions; does not establish forecast or adjustment.

Source basis: Back.

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Appendix: why the three questions are ranked

1. **Margin and run-rate.** The utilization/margin tension leaves some stones unturned. The answer could alter margin forecasts and valuation.
2. **Nature and duration of commissioning/quality costs.** Identifiable, finite, incremental and non-run-rate evidence affects whether any normalization is supportable.
3. **Breadth and durability of volume recovery.** Evidence across customer, channel, end market and facility can change forecast confidence, risk and peer selection.

Acceptable alternative

A question about qualification timing or direct-account/distributor economics can be equally strong if it states a potential valuation consequence and is prioritized against these questions.

Source basis: Back.

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Appendix: My prompt-chain skeleton

Message 1: evidence / analysis

Use only the supplied TargetCo material. Extract business-model evidence; recompute revenue growth; distinguish disclosed volume and price/mix contribution points; track utilization and margin; identify tensions; return source location, support and limit. No valuation or final prose.

Message 2: challenge + draft

Check the analysis against the four criteria. Correct unsupported inference, omitted dimensions, numerical inconsistency and generic questions. Then draft the one-page note under Business model; Historical trajectory; Key tensions; Three ranked questions + why.

Message 3: verify + revise

Reconcile each material number and statement to source or calculation. Revise the note only for verified corrections. Flag remaining uncertainty rather than supplying a fact.

Source basis: Back.

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Appendix: a strong near-final prompt sequence for TargetCo

Message 1 – evidence / analysis. The supplied TargetCo profile is authoritative. Use no external facts, tools or valuation. Extract product/qualification, markets, channel and footprint; revenue and $g_t = R_t/R_{t-1} - 1$; disclosed volume and price/mix contribution points (check $g_t = v_t + p_t$); utilization and reported margin; and M-01/M-02 as management statements. For each, give profile location, classification, calculation/support and limit. Flag 2025A's 84% versus 82% utilization but 14.3% versus 14.8% margin tension. Do not draft.

Message 2 – challenge + draft. Assess the extraction against four criteria: coherent business-model synthesis; correct contribution arithmetic and qualified causality; evidence/uncertainty discipline; concise VP usefulness with exactly three ranked questions naming a later judgement. Correct only what the profile supports. Draft one page: Business model / positioning; Historical trajectory; Key tensions; Three ranked questions + why. Preserve M-01/M-02 as management statements. No valuation.

Message 3 – verify + revise. Reconcile every material number and statement to a profile location or calculation. Recheck 2023A ($-4.0 \text{ ppt} + 6.1 \text{ ppt} = 2.1\%$) and 2025A ($4.0 \text{ ppt} + 1.9 \text{ ppt} = 5.9\%$). Confirm that no management statement became fact, causal language remains qualified and exactly three questions are ranked. Give a short reconciliation list, revise only where required, and flag unresolved uncertainty.

Source basis: Back.

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Appendix: common weak moves are diagnostic

Weak move	Why it fails and how to repair it
Rewrites profile headings	It has not synthesized an operating configuration. Connect product, qualification, channel and geography cautiously.
Lists annual data	It misses changing economics. Contrast 2023A price/mix contribution offsetting falling volume with 2025A's larger volume contribution.
Says utilization "caused" margin recovery	The pack has no quantified causal bridge. Use qualified language and ask for the bridge.
Calls M-01/M-02 facts	It converts management language into established evidence. Label the statement and specify missing support.
Calls 9.4% / 41.8% high or low	Benchmark and contractual context are absent. State the figures and ask for context.
Asks generic diligence questions	A VP cannot see valuation relevance. Name the later judgement that might change.

Source basis: Back.

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Appendix: Session 1 and the AI Fluency 4Ds

4D	Session 1 action	Evidence retained
Delegation	Delegate extraction, cross-check suggestions and drafting; retain authority, valuation relevance and approval.	Work Record: delegated versus retained.
Description	State task, authoritative source, output, criteria and checks.	P0 and P1–P4 versions.
Discernment	Score four dimensions; identify unsupported stories and omissions.	Rubric, challenge, baseline/redo.
Diligence	Reconcile sources; disclose uncertainty; do not treat management language as verified.	Matrix, numerical check, remaining uncertainty.

Mode

This is augmentation: AI extends the analysis.

Source basis: Back.

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Appendix: References

Rosenbaum & Pearl (2013), *Investment Banking*

Ch. 1, Step I “Study the Target”; Exhibit 1.3 Business and Financial Profile Framework; “Financial Profile”; and “Calculation of Key Financial Statistics and Ratios.”

Rosenbaum & Pearl Workbook (2013) and Koller et al. (2025)

Workbook Ch. 1 Q14–21 (business/financial profile), Q7–9 (growth/profitability/credit calculations), Q3 (one-time/non-recurring adjustments); Koller Chs. 11 and 13.

Source basis: Back.

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